

TWO INTEGRAL DEGENERATION TYPES OF THE SIGNED ERDŐS–STRAUS QUARTIC

THE CLANKERS

ABSTRACT. We specialize the finite signed completion of the literal-root ES quartic at its original prime. Its integral normalization index is p^2 for Type I and p^9 for Type II, despite constant rank eight. The original coordinate conductors, Smith factors and generic rational-point counts are explicit. The new Type-I boundary point lifts modulo p^2 but never modulo p^3 . These statements compare original arithmetic states; they do not prove an ES state exists at every prime.

1. THE ORIGINAL SOURCE AND ITS FINITE SIGNED COMPLETION

Let $p \equiv 1 \pmod{12}$ be prime. Retain $p/4 < a < p/2$, $R = 4a - p$, $u \mid a^2$, and either $E : R \mid 4u + 1$ or $M : R \mid u + a$. With

$$d = \gcd(a, u), \quad (h, r, s) = (d^2/u, u/d, a/d),$$

the original Type-I/II denominators are

$$E : (a, hsk, phrk), \quad \kappa = (pr + s)/R; \quad M : (a, phs\lambda, phr\lambda), \quad \lambda = (r + s)/R.$$

These inherited coordinates [1, §2] retain $a = hrs$, $u = hr^2$ and $\gcd(r, s) = 1$. The ordered inverses are $a^2/(Ry - pa)$ and $pa^2/(Ry - pa)$. Reciprocity gives $(l/R) = (l/p)$ for every prime $l \mid a$, including two by the supplementary law. Hence

$$(1) \quad E : (h/p) = -1, \quad M : (rs/p) = -1.$$

The M claim uses $r \equiv -s \pmod{R}$.

Set $\mathfrak{o} = \mathbb{Z}_p$, $K = \mathbb{Q}_p$, and retain the literal ordered roots

$$(\ell_0, \ell_1, \ell_2, \ell_3) = (p, x, y, z), \quad S = p + x + y + z, \quad H(T) = -S^{-1} \prod_i (T - \ell_i).$$

The cubic coefficient of H is one. The given signed-completion construction [2, §3, (18)–(24)] is

$$(2) \quad \widehat{S} = \mathfrak{o}[T, \eta]/(H, \eta^2 - H'),$$

with original basis $1, T, T^2, T^3, \eta, \eta T, \eta T^2, \eta T^3$. It recovers the original affine inverse only on $\eta \neq 0$, with $\xi = \eta^{-1}$. We prove below that S is a p -unit, so (2) is indeed finite free of rank eight over this original arithmetic ring.

For E write $(x, y, z) = (a, b, pc)$. The original identity $pD + 1 = 4hr\kappa$ gives $c \equiv 1/4 \pmod{p}$. Both $a - b = hs(r - \kappa)$ and $a + b = hs(r + \kappa)$ are units: the contrary would make the nonresidue hr^2 equal to $1/4$ or $-1/4$ modulo p . Thus only ℓ_0, ℓ_3 have a nonunit difference, of valuation one, and $S \equiv a + b$.

For M write $(x, y, z) = (a, pb, pc)$. Here b, c are units, $4bc \equiv b + c$, and $(bc/p) = -1$ by (1). They are distinct modulo p because $r, s < p$ are distinct. Neither is one: then the other would be $1/3$, contrary to $(3/p) = 1$. Thus the cluster ℓ_0, ℓ_2, ℓ_3 has all pair differences of valuation one, its singleton a is separated by units, and $S \equiv a$.

Writing $d_i = H'(\ell_i)$, we have therefore proved

$$(3) \quad E : (v_p(d_i)) = (1, 0, 0, 1), \quad M : (v_p(d_i)) = (2, 0, 2, 2).$$

2. COMPLETE NORMALIZATION AND CONDUCTOR

Let $k_i = \lfloor v_p(d_i)/2 \rfloor$ and

$$N_i = \mathfrak{o}[\theta_i]/(\theta_i^2 - d_i/p^{2k_i}), \quad N = \prod_i N_i.$$

Each N_i is maximal: its quadratic is either a split unit quadratic, an unramified unit quadratic, or an Eisenstein quadratic of odd residue characteristic. A direct proof is obtained by taking trace and norm of $a + b\theta_i$: trace forces $a \in \mathfrak{o}$, then norm forces $b \in \mathfrak{o}$. In the split case use the two values and the unit $2\sqrt{d_i/p^{2k_i}}$. The exact normalization map is

$$(4) \quad f(T) + \eta g(T) \mapsto (f(\ell_i) + p^{k_i}g(\ell_i)\theta_i)_i.$$

Theorem 2.1. *For Type I,*

$$N/\widehat{S} \simeq (\mathfrak{o}/p)^2, \quad \mathfrak{f} = pN_0 \times N_1 \times N_2 \times pN_3.$$

For Type II,

$$N/\widehat{S} \simeq (\mathfrak{o}/p)^2 \oplus (\mathfrak{o}/p^2)^2 \oplus \mathfrak{o}/p^3, \quad \mathfrak{f} = p^3N_0 \times N_1 \times p^3N_2 \times p^3N_3.$$

Thus the normalization indices are p^2 and p^9 , respectively.

Proof. Unit resultants split off the noncolliding root factors integrally. For E the remaining root lattice is $L_2 = \{(v_0, v_3) : v_0 \equiv v_3 \pmod{p}\}$, by linear interpolation across a root difference of exact valuation one. Its Smith factors are $1, p$. Equation (4) gives $L_2 \oplus L_2$, and four already maximal outside coordinates. A normal value supported at one cluster root belongs to the order precisely when both its coefficients in $1, \theta_i$ are divisible by p . This gives the quotient and maximal conductor ideal in the Type-I case.

For M use the Newton basis $1, T - p, (T - p)(T - pb)$ on the three clustered roots. Evaluation has Smith factors $1, p, p^2$, so its root lattice L_3 has index p^3 . Equation (4) gives $L_3 \oplus pL_3$, with exponents $(0, 1, 2, 1, 2, 3)$, and two outside zeros. A value $a + b\theta_i$ supported at one cluster root has Lagrange denominators of order p^2 . Thus membership requires $a \in p^2\mathfrak{o}, b \in p^3\mathfrak{o}$. Membership after multiplication by the unit θ_i forces both into $p^3\mathfrak{o}$. That condition is also sufficient, proving the Type-II conductor and quotient. $\square$

In particular the eight Smith exponents are

$$(0, 0, 0, 0, 0, 0, 1, 1) \quad \text{and} \quad (0, 0, 0, 1, 1, 2, 2, 3).$$

The full inverse of (4) interpolates the values a_i and b_i/p^{k_i} separately. It is integral exactly when all eight original polynomial coefficients are integral. The nonzero finite quotient is retained; no $\mathfrak{o}$ -linear section from that torsion quotient into the torsion-free N exists. These conductors are conductors of finite orders, not the analytic weighted conductor used elsewhere in the programme.

3. GENERIC SIGNED POINTS AND SPECIAL BOUNDARY POINTS

Over K , (2) is $\prod_i K[\eta_i]/(\eta_i^2 - d_i)$. For E the two valuation-one square classes are equal, because their unit ratio is $-1 \pmod{p}$. At the other two roots the derivative residues are $-a^2(a - b)/(a + b)$ and $-b^2(b - a)/(a + b)$, also with square ratio. Thus the fibre has either four K -points or none; its nonsplit factors are two copies of a ramified quadratic and possibly two copies of an unramified quadratic. For M, the singleton derivative is $-a^2 \pmod{p}$. The three cluster derivatives after division by p^2 have residues

$$(1 - b)(1 - c), \quad (b - 1)(b - c), \quad (c - 1)(c - b).$$

Their product is a negative square. Either all three or exactly one is square, so the complete fibre has eight or four K -points. Unit square roots lift by $z_{n+1} = z_n - p^n((z_n^2 - d)/p^n)(2z_n)^{-1} \pmod{p^{n+1}}$. This accounts for every signed point and every field factor.

The colliding special blocks are respectively

$$\mathbb{F}_p[\eta]/\eta^4, \quad \mathbb{F}_p[\epsilon, \eta]/(\epsilon^3, \eta^2 - 3\epsilon^2).$$

Their lengths are four and six, while multiplication by η has Jordan blocks (4) and (4, 2). The original inverse obtained by inverting η deletes these entire nonzero local blocks; it does not make their poles finite.

Theorem 3.1. *The added Type-I point $(T, \eta) = (0, 0) \pmod{p}$ has exactly p lifts modulo p^2 , but no lift modulo p^3 .*

Proof. Factor $H(T) = U(T)(T - p)(T - pc)$, where $U(0) = -ab/S$ is a unit and $c = 1/4 \pmod{p}$. For $T = pw, \eta = pv$, the root equation is automatic modulo p^2 , and the quadratic equation requires $2w - 1 - c = 0 \pmod{p}$. Thus $w = 5/8$ and all p choices of v work. But every next lift has

$$H(T)/p^2 \equiv -U(0)(1 - c)^2/4 = 9ab/(64S) \not\equiv 0 \pmod{p}.$$

The root equation fails already modulo p^3 . □

At the hard prime 1009, the original E denominators (253, 87032, 3818056) have *no* K -rational signed inverse, while the original M denominators (253, 2042216, 88792) have eight. Both are positive integer ES identities. Thus degree-eight finite flatness is not rational-point or integral-section surjectivity. The proper map retains length at its boundary; it does not discharge the prime-by-prime original occupancy problem.

The distinguished-prime idempotent

$$e_p(T) = \frac{\prod_{i=1}^3 (T - \ell_i)}{\prod_{i=1}^3 (p - \ell_i)}$$

has an exact denominator p in E and p^2 in M. Hence the generic prime-only sign map $\eta \mapsto (1 - 2e_p)\eta$ does not preserve the original integral order, whereas the global sign does. Both extend to the normalization. This states the actual loss instead of deleting the original prime-root marking.

4. PROVENANCE AND EXECUTION BOUNDARY

The literal target, finite rank-eight extension and open inverse are supplied antecedents [2]. The indices, original conductors, local signed types, and sharp finite lifting obstruction are derived here. No historical-priority claim is made. The full companion proof gives all coordinate maps, the reciprocal quadratic twist, and explicit finite-place rational controls satisfying the supplied height inequalities but missing one global denominator condition. It also transports the previously proved $p = 825241$ capacity crossing; that crossing is supplied by original divisor arithmetic, not by assuming an inverse point of this finite cover.

The package's standard-library verifier and a separate primitive-source checker compare every labelled E/M state at all primes $p \equiv 1 \pmod{12}$ through 3000. The detailed original lattice and inverse examples, finite special fibres, negative controls and command receipts are included. Finite replay does not prove initial ES occupancy, a Lean build or an independent mathematical review.

REFERENCES

- [1] C. Elsholtz and T. Tao (2013), *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Aust. Math. Soc. 94, 50–105, §2. <https://arxiv.org/abs/1107.1010>.
- [2] The Clankers / Split-Zero continuation (2026), supplied `Pasted markdown(20260920-025519).md`, §§1–3, especially equations (8)–(11) and (18)–(24). Original source preserved with this package.